Marcus W. Fedarko

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Education

Ph.D., Computer Science — 9/2018–6/2025
University of California, San Diego — La Jolla, CA

M.S., Computer Science — 9/2018–6/2022
University of California, San Diego — La Jolla, CA

B.S. with High Honors, Computer Science — 9/2014–5/2018
University of Maryland — College Park, MD

Manuscripts in Preparation

2. **Fedarko MW**, Zhang Z, Bankevich A, and Pevzner PA. "Metagenome assembly of long and accurate reads using iterative downsampling."

1. Zhang Z, **Fedarko MW**, Bankevich A, and Pevzner PA. "Consensus assembly of long and accurate reads."

Refereed Publications

7. **Fedarko MW**, Kolmogorov M, and Pevzner PA (2022). "Analyzing rare mutations in metagenomes assembled using long and accurate reads." *Genome Research, 32*(11-12):2119–2133.

6. Cantrell K*, **Fedarko MW***, Rahman G, McDonald D, Yang Y, Zaw T, Gonzalez A, Janssen S, Estaki M, Haiminen N, Beck KL, Zhu Q, Sayyari E, Morton JT, Armstrong G, Tripathi A, Gauglitz JM, Marotz C, Matteson NL, Martino C, Sanders JG, Carrieri AP, Song SJ, Swafford AD, Dorrestein PC, Andersen KG, Parida L, Kim H-C, Vázquez-Baeza Y, and Knight R (2021). "EMPress Enables Tree-Guided, Interactive, and Exploratory Analyses of Multi-omic Data Sets." *mSystems, 6*(2):e01216-20. (* = contributed equally)

5. Huey SL, Jiang L, **Fedarko MW**, McDonald D, Martino C, Ali F, Russell DG, Udipi SA, Thorat A, Thakker V, Ghugre P, Potdar RD, Chopra H, Rajagopalan K, Haas JD, Finkelstein JL, Knight R, and Mehta S (2020). "Nutrition and the Gut Microbiota in 10- to 18-Month-Old Children Living in Urban Slums of Mumbai, India." *mSphere, 5*(5):e00731-20.

4. **Fedarko MW**, Martino C, Morton JT, González A, Rahman G, Marotz CA, Minich JJ, Allen EA, and Knight R (2020). "Visualizing 'omic feature rankings and log-ratios using Qurro." *NAR Genomics and Bioinformatics, 2*(2):lqaa023.

3. Sanders JG, Nurk S, Salido RA, Minich J, Xu ZZ, Martino C, **Fedarko M**, Arthur TD, Chen F, Boland BS, Humphrey GC, Brennan C, Sanders K, Gaffney J, Jepsen K, Khosroheidari M, Green C, Liyange M, Dang JW, Phelan VV, Quinn RA, Bankevich A, Chang JT, Rana TM, Conrad DJ, Sandborn WJ, Smarr L, Dorrestein PC, Pevzner PA, and Knight R (2019). "Optimizing sequencing protocols for leaderboard metagenomics by combining long and short reads." *Genome Biology, 20*(1):226.

2. Ghurye J, Treangen T, **Fedarko M**, Hervey WJ, and Pop M (2019). "MetaCarvel: linking assembly graph motifs to biological variants." *Genome Biology, 20*(1):174.

1. Meisel JS, Nasko DJ, Brubach B, Cepeda-Espinoza V, Chopyk J, Corrada-Bravo H, **Fedarko M**,

Ghurye J, Javkar K, Olson ND, Shah N, Allard SM, Bazinet AL, Bergman NH, Brown A, Caporaso JG, Conlan S, DiRuggiero J, Forry SP, Hasan NA, Kralj J, Luethy PM, Milton DK, Ondov BD, Preheim S, Ratnayake S, Rogers SM, Rosovitz MJ, Sakowski EG, Schliebs NO, Sommer DD, Ternus KL, Uritskiy G, Zhang SX, Pop M, and Treangen TJ (2018). "Current progress and future opportunities in applications of bioinformatics for biodefense and pathogen detection: Report from the Winter Mid-Atlantic Microbiome Meet-up, College Park, MD January 10th, 2018." *Microbiome, 6*(1):197.

Open-Source Software

6. **wotplot**: Library for creating and visualizing dot plot matrices.
   https://github.com/fedarko/wotplot
5. **strainFlye** (☕$_7$): Pipeline for the analysis of rare mutations in metagenomes.
   https://github.com/fedarko/strainFlye
4. **EMPress** (☕$_6$): Visualization tool for phylogenetic trees and associated data.
   https://github.com/biocore/empress
3. **Qurro** (☕$_4$): Visualization tool for log-ratios of compositional data.
   https://github.com/biocore/qurro
2. **pyfastg**: Library for parsing SPAdes FASTG files.
   https://github.com/fedarko/pyfastg
1. **MetagenomeScope**: Visualization tool for metagenome assembly graphs.
   https://github.com/marbl/MetagenomeScope

Projects marked (☕$_n$) represent the "main contribution" of the n-th refereed publication in the list above.

Talks

4. "Metagenome assembly." 1/2024
   Guest lecture for ESE 184 (Computational Tools for Decoding Microbial Ecosystems), California Institute of Technology.
3. "Studying microbiomes using DNA sequencing." 5/2023
   Presentation to students from Kearny High School visiting UC San Diego.
2. "Visualizing, Exploring, and Understanding Microbiome Sequencing Data." 1/2020
   UC San Diego CSE Research Open House.
1. "Visualizing Metagenomic Assembly Graphs, Doing Undergrad Research at UMD, Applying to Grad Schools, and probably other stuff along the way." 4/2018
   Guest lecture for CMSC 396H (Undergraduate Honors Seminar), University of Maryland.

Editing and Reviewing

2. Textbook Editor, *Advanced Bioinformatics Algorithms* 2024
1. Peer Reviewer, *PLOS ONE* 2023

Service

5. System Administrator, Pevzner Lab computing server 2/2023–Present
4. Mentor, UC San Diego Graduate Women in Computing mentorship program 10/2021–6/2024

3. Moderator, QIIME 2 forum (https://forum.qiime2.org) 3/2020–4/2024
2. Co-organizer, UC San Diego CSE Visit Day 1/2019–3/2024
1. Code Review (Co-)organizer, Knight Lab 12/2018–8/2020

Research Experience

Graduate Student Researcher 9/2018–Present
University of California, San Diego La Jolla, CA

- Designing visualization and metagenome assembly methods for high-throughput microbiome sequencing data.
- Assisting with various software and analysis projects.

Research Intern 6/2016–8/2018
University of Maryland College Park, MD

- Designed MetagenomeScope, a visualization tool for metagenome assembly graphs.

Teaching Experience

Teaching Assistant
University of California, San Diego La Jolla, CA

- **CSE 185**: Advanced Bioinformatics Laboratory 3/2025–6/2025
- **CSE 181**: Molecular Sequence Analysis 1/2025–3/2025
  1/2024–3/2024
- **CSE 282**: Introduction to Bioinformatics Algorithms 1/2025–3/2025
  1/2023–3/2023
  1/2022–3/2022
  1/2021–3/2021

Course Assistant
Marine Biological Laboratory Woods Hole, MA

- **STAMPS**: Strategies and Techniques for Analyzing Microbial Population Structures 7/2018–8/2018
- **MOLE**: Workshop on Molecular Evolution 7/2018

Teaching Assistant
University of Maryland College Park, MD

- **CMSC 330**: Organization of Programming Languages 8/2016–12/2016

Professional Experience

Software Developer 7/2025–Present
***Bioinformatics Algorithms* (Compeau & Pevzner)** Remote

- Implementing and editing automatic code challenges for a textbook.

Student Staff Writer 1/2015–9/2017
University of Maryland Department of Computer Science College Park, MD

- Wrote and edited articles for the department's website and other media.

- Assisted with the logistics of various department outreach events.

Student Intern — 5/2013–8/2014
Axiometric — Columbia, MD

- Designed a graphical interface to an RF propagation model to assist clients in planning deployments of mesh networks of utility meters.
- Aided in the creation and maintenance of other utility meter deployment management software.

Intern Software Engineer — 7/2012–8/2012
Battlefield Telecommunications Systems — Columbia, MD

- Designed a web interface to monitor the connection strength of radio devices.
- Helped integrate this functionality into the company's existing network management user interface.

Honors and Awards

10. **UC San Diego:** Chez Bob Hackathon Award — 2024
9. **University of Maryland:** CMNS Dean's List — 2014–2018
8. **University of Maryland:** University Honors Citation — 2017
7. **University of Maryland:** Rita Colwell Travel Fellowship — 2017
6. **University of Michigan:** Travel Award, EGS Workshop — 2017
5. **University of Maryland:** John D. Gannon Endowed Scholarship — 2017
4. **University of Maryland:** Corporate Partners in Computing Scholarship — 2016, 2017
3. **Omicron Delta Kappa National Leadership Honor Society** — 2016
2. **Northrop Grumman:** Scholarship for Employees' Children — 2014
1. **University of Maryland:** Dean's Scholarship — 2014

Skills

- **Languages:** English (native), German (basic), Hindi (basic)
- **Computer Languages:** Python, JavaScript, HTML / CSS, LaTeX, Java, C, C++, ...
- **Software Libraries:** matplotlib, pandas, NumPy, SciPy, scikit-bio, jQuery, Cytoscape.js, ...